

G1 Understanding differentiation

Name: _____

Class: _____

Date: _____

Time: **428 minutes**

Marks: **357 marks**

Comments:

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Q1.

Craig is investigating the gradient of chords of the curve with equation $f(x) = x - x^2$

Each chord joins the point $(3, -6)$ to the point $(3 + h, f(3 + h))$

The table shows some of Craig's results.

x	$f(x)$	h	$x + h$	$f(x + h)$	Gradient
3	-6	1	4	-12	-6
3	-6	0.1	3.1	-6.51	-5.1
3	-6	0.01			
3	-6	0.001			
3	-6	0.0001			

- (a) Show how the value -5.1 has been calculated. (1)
- (b) Complete the third row of the table above. (2)
- (c) State the limit suggested by Craig's investigation for the gradient of these chords as h tends to 0 (1)
- (d) Using differentiation from first principles, verify that your result in part (c) is correct. (4)
- (Total 8 marks)**

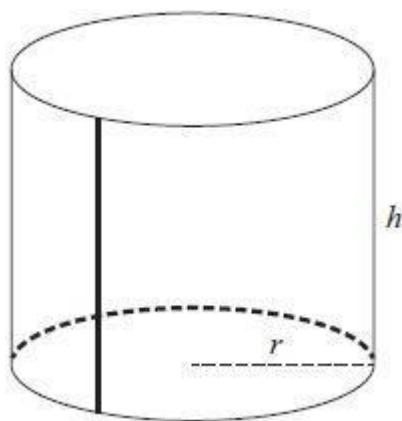
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Q2.

Rakti makes open-topped cylindrical planters out of thin sheets of galvanised steel.

She bends a rectangle of steel to make an open cylinder and welds the joint. She then welds this cylinder to the circumference of a circular base.



The planter must have a capacity of 8000 cm^3

Welding is time consuming, so Rakti wants the total length of weld to be a minimum.

Calculate the radius, r , and height, h , of a planter which requires the minimum total length of weld.

Fully justify your answers, giving them to an appropriate degree of accuracy.

(Total 9 marks)

Q3.

A curve has equation $y = x^3 - 48x$

The point A on the curve has x coordinate -4

The point B on the curve has x coordinate $-4 + h$

(a) Show that the gradient of the line AB is $h^2 - 12h$

(4)

(b) Explain how the result of part (a) can be used to show that A is a stationary point on the curve.

(2)

(Total 6 marks)

Q4.

A function f is defined by $f(x) = \frac{x}{\sqrt{2x-2}}$

(a) State the maximum possible domain of f .

(2)

(b) Use the quotient rule to show that $f'(x) = \frac{x-2}{(2x-2)^{\frac{3}{2}}}$

(3)

- (c) Show that the graph of $y = f(x)$ has exactly one point of inflection.

(7)

- (d) Write down the values of x for which the graph of $y = f(x)$ is convex.

(1)

(Total 13 marks)

Q5.

- (a) Given that $f(x) = x^2 - 4x + 2$, find $f(3 + h)$

Express your answer in the form $h^2 + bh + c$, where b and $c \in \mathbb{Z}$.

(2)

- (b) The curve with equation $y = x^2 - 4x + 2$ passes through the point $P(3, -1)$ and the point Q where $x = 3 + h$.

Using differentiation from first principles, find the gradient of the tangent to the curve at the point P .

(3)

(Total 5 marks)

Q6.

A curve has equation $y = \frac{2x+3}{4x^2+7}$

- (a) (i) Find $\frac{dy}{dx}$

(2)

- (ii) Hence show that y is increasing when $4x^2 + 12x - 7 < 0$

(4)

- (b) Find the values of x for which y is increasing.

(2)

(Total 8 marks)

Q7.

An open-topped fish tank is to be made for an aquarium.

It will have a square horizontal base, rectangular vertical sides and a volume of 60 m^3

The materials cost:

- £15 per m^2 for the base
- £8 per m^2 for the sides.

- (a) Modelling the sides and base of the fish tank as laminae, use calculus to find the height of the tank for which the overall cost of the materials has its minimum value.

Fully justify your answer.

(8)

- (b) (i) **In reality, the thickness of the base and sides of the tank is 2.5 cm**

Briefly explain how you would refine your modelling to take account of the thickness of the sides and base of the tank.

(1)

- (ii) How would your refinement affect your answer to part (a)?

(1)

(Total 10 marks)

Q8.

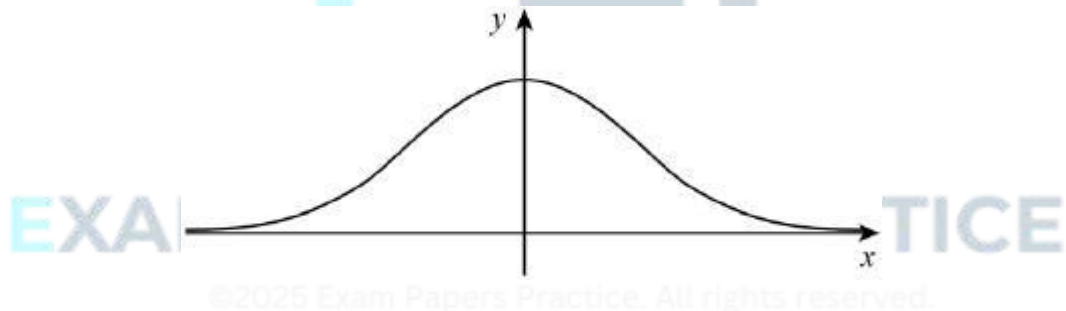
$$f(x) = \sin x$$

Using differentiation from first principles find the exact value of $f'\left(\frac{\pi}{6}\right)$
 Fully justify your answer.

(Total 6 marks)

Q9.

The diagram shows part of the graph of $y = e^{-x^2}$

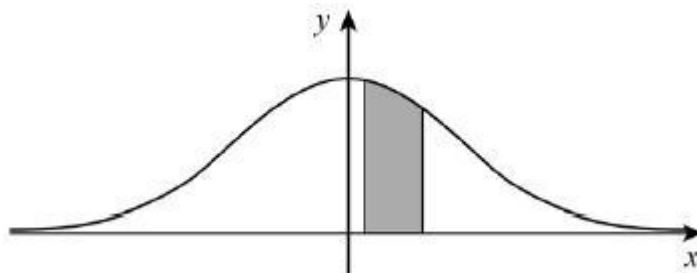


The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

- (a) Find the values of x for which the graph is concave.

(4)

- (b) The finite region bounded by the x -axis and the lines $x = 0.1$ and $x = 0.5$ is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2} dx$

Give your estimate to four decimal places.

(3)

- (c) Explain with reference to your answer in part (a), why the answer you found in part (b) is an underestimate.

(2)

- (d) By considering the area of a rectangle, and using your answer to part (b), prove that the shaded area is 0.4 correct to 1 decimal place.

(3)

(Total 12 marks)

Q10.

A bird flies from a tree. At time t seconds, the bird's height, y metres, above the horizontal ground is given by

$$y = \frac{1}{8}t^4 - t^2 + 5, \quad 0 \leq t \leq 4$$

- (a) Find $\frac{dy}{dt}$.

(2)

- (b) (i) Find the rate of change of height of the bird in metres per second when $t = 1$.

(2)

- (ii) Determine, with a reason, whether the bird's height above the horizontal ground is increasing or decreasing when $t = 1$.

(1)

- (c) (i) Find the value of $\frac{d^2y}{dt^2}$ when $t = 2$.

(2)

- (ii) Given that y has a stationary value when $t = 2$, state whether this is a maximum value or a minimum value.

(1)

(Total 8 marks)

Q11.

The gradient, $\frac{dy}{dx}$, of a curve at the point (x, y) is given by

$$\frac{dy}{dx} = 10x^4 - 6x^2 + 5$$

The curve passes through the point $P(1, 4)$.

- (a) Find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$.

(3)

- (b) Find the equation of the curve.

(5)

(Total 8 marks)

Q12.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 - 4x + 15$.

- (i) Use the Factor Theorem to show that $x + 3$ is a factor of $f(x)$.

(2)

- (ii) Express $f(x)$ in the form $(x + 3)(x^2 + px + q)$, where p and q are integers.

(2)

- (b) A curve has equation $y = x^4 - 8x^2 + 60x + 7$.

- (i) Find $\frac{dy}{dx}$.

(3)

- (ii) Show that the x -coordinates of any stationary points of the curve satisfy the equation

$$x^3 - 4x + 15 = 0$$

(1)

- (iii) Use the results above to show that the only stationary point of the curve occurs when $x = -3$.

(2)

- (iv) Find the value of $\frac{d^2y}{dx^2}$ when $x = -3$.

(3)

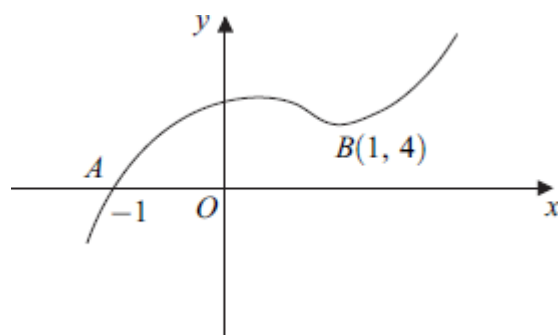
- (v) Hence determine, with a reason, whether the curve has a maximum point or a minimum point when $x = -3$.

(1)

(Total 14 marks)

Q13.

The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(a) Given that $y = x^5 - 3x^2 + x + 5$, find:

(i) $\frac{dy}{dx}$;

(3)

(ii) $\frac{d^2y}{dx^2}$.

(1)

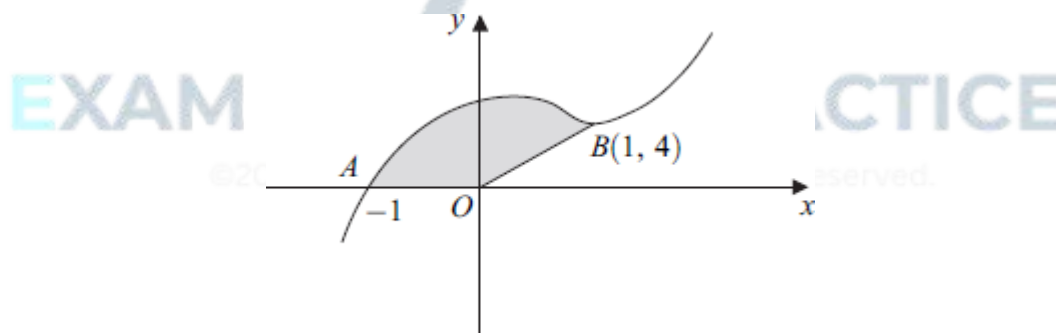
(b) Find an equation of the tangent to the curve at the point $A(-1, 0)$.

(2)

(c) Verify that the point B , where $x = 1$, is a minimum point of the curve.

(3)

(d) The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(i) Find $\int_{-1}^1 (x^5 - 3x^2 + x + 5) dx$.

(5)

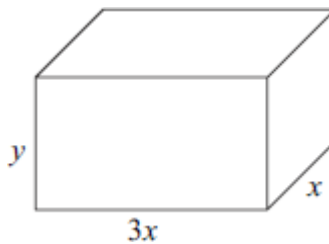
(ii) Hence find the area of the shaded region bounded by the curve between A and B and the line segments AO and OB .

(2)

(Total 16 marks)

Q14.

The diagram shows a solid cuboid with sides of lengths x cm, $3x$ cm and y cm.



The total surface area of the cuboid is 32 cm^2 .

- (a) (i) Show that $3x^2 + 4xy = 16$.

(2)

- (ii) Hence show that the volume, $V \text{ cm}^3$, of the cuboid is given by

$$V = 12x - \frac{9x^3}{4}$$

(2)

- (b) Find $\frac{dV}{dx}$.

(2)

- (c) (i) Verify that a stationary value of V occurs when $x = \frac{4}{3}$.

(2)

- (ii) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = \frac{4}{3}$.

(2)

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(Total 10 marks)

Q15.

The gradient, $\frac{dy}{dx}$, of a curve C at the point (x, y) is given by

$$\frac{dy}{dx} = 20x - 6x^2 - 16$$

- (a) (i) Show that y is increasing when $3x^2 - 10x + 8 < 0$.

(2)

- (ii) Solve the inequality $3x^2 - 10x + 8 < 0$.

(4)

- (b) The curve C passes through the point $P(2, 3)$.

- (i) Verify that the tangent to the curve at P is parallel to the x-axis. (2)
- (ii) The point $Q(3, -1)$ also lies on the curve. The normal to the curve at Q and the tangent to the curve at P intersect at the point R . Find the coordinates of R . (7)
- (Total 15 marks)

Q16.

At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^2 - \frac{4}{x^2} - 11$$

The point $P(2, 1)$ lies on the curve.

- (a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 2$. (1)
- (ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 2$. (4)
- (iii) Hence state whether P is a maximum point or a minimum point, giving a reason for your answer. (1)
- (b) Find the equation of the curve. (4)
- (Total 10 marks)

Q17.

- (a) Given that $x = \frac{\sin y}{\cos y}$, use the quotient rule to show that

$$\frac{dx}{dy} = \sec^2 y$$

(3)

- (b) Given that $\tan y = x - 1$, use a trigonometrical identity to show that

$$\sec^2 y = x^2 - 2x + 2$$

(2)

- (c) Show that, if $y = \tan^{-1}(x - 1)$, then

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

(1)

(d) A curve has equation $y = \tan^{-1}(x - 1) - \ln x$.

(i) Find the value of the x -coordinate of each of the stationary points of the curve.

(4)

(ii) Find $\frac{d^2y}{dx^2}$.

(2)

(iii) Hence show that the curve has a minimum point which lies on the x -axis.

(2)

(Total 14 marks)

Q18.

A curve has equation $y = x^4 + x$.

(a) Find the gradient of the line passing through the point $(-2, 14)$ and the point on the curve for which $x = -2 + h$. Give your answer in the form

$$p + qh + rh^2 + h^3$$

where p , q and r are integers.

(5)

(b) Show how the answer to part (a) can be used to find the gradient of the curve at the point $(-2, 14)$. State the value of this gradient.

(2)

(Total 7 marks)

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Q19.

The curve with equation $y = 13 + 18x + 3x^2 - 4x^3$ passes through the point P where $x = -1$.

(a) Find $\frac{dy}{dx}$

(3)

(b) Show that the point P is a stationary point of the curve and find the other value of x where the curve has a stationary point

(3)

(c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P .

(3)

- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)

(Total 10 marks)

Q20.

The volume, $V \text{ m}^3$, of water in a tank after time t seconds is given by

$$V = \frac{t^3}{4} - 3t + 5$$

- (a) Find $\frac{dV}{dt}$.

(2)

- (b) (i) Find the rate of change of volume, in $\text{m}^3 \text{ s}^{-1}$, when $t = 1$.

(2)

- (ii) Hence determine, with a reason, whether the volume is increasing or decreasing when $t = 1$.

(1)

- (c) (i) Find the positive value of t for which V has a stationary value.

(3)

- (ii) Find $\frac{d^2V}{dt^2}$, and hence determine whether this stationary value is a maximum value or a minimum value.

(3)

(Total 11 marks)

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Q21.

- (a) Expand $(5 + h)^3$.

(1)

- (b) A curve has equation $y = x^3 - x^2$.

- (i) Find the gradient of the line passing through the point $(5, 100)$ and the point on the curve for which $x = 5 + h$. Give your answer in the form

$$p + qh + rh^2$$

where p , q and r are integers.

(4)

- (ii) Show how the answer to part (b)(i) can be used to find the gradient of the curve at the point (5, 100). State the value of this gradient.

(2)

(Total 7 marks)

Q22.

The depth of water, y metres, in a tank after time t hours is given by

$$y = \frac{1}{8}t^4 - 2t^2 + 4t, \quad 0 \leq t \leq 4$$

- (a) Find:

(i) $\frac{dy}{dt}$,

(3)

(ii) $\frac{d^2y}{dt^2}$.

(2)

- (b) Verify that y has a stationary value when $t = 2$ and determine whether it is a maximum value or a minimum value.

(4)

- (c) (i) Find the rate of change of the depth of water, in metres per hour, when $t = 1$.

(2)

- (ii) Hence determine, with a reason, whether the depth of water is increasing or decreasing when $t = 1$.

(1)

(Total 12 marks)

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Q23.

It is given that $y = \tan 4x$.

- (a) By writing $\tan 4x$ as $\frac{\sin 4x}{\cos 4x}$, use the quotient rule to show that

$$\frac{dy}{dx} = p(1 + \tan^2 4x),$$

where p is a number to be determined.

(3)

- (b) Show that $\frac{d^2y}{dx^2} = q(1 + y^2)$, where q is a number to be determined.

(5)

(Total 8 marks)

Q24.

A botanist is investigating the rate of growth of a certain species of toadstool. She observes that a particular toadstool of this type has a height of 57 millimetres at a time 12 hours after it begins to grow.

$$h = A \left(1 - e^{-\frac{1}{4}t} \right)$$

She proposes the model $h = A \left(1 - e^{-\frac{1}{4}t} \right)$, where A is a constant, for the height h millimetres of the toadstool, t hours after it begins to grow.

(a) Use this model to:

(i) find the height of the toadstool when $t = 0$;

(1)

(ii) show that $A = 60$, correct to two significant figures.

(2)

$$h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$$

(b) Use the model $h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$ to:

(i) show that the time T hours for the toadstool to grow to a height of 48 millimetres is given by

$$T = a \ln b$$

where a and b are integers;

(3)

(ii) show that $\frac{dh}{dt} = 15 - \frac{h}{4}$;

(3)

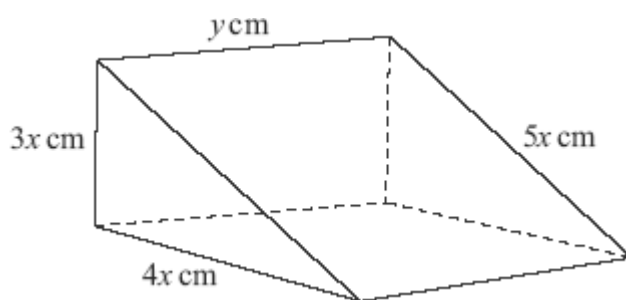
(iii) find the height of the toadstool when it is growing at a rate of 13 millimetres per hour.

(1)

(Total 10 marks)

Q25.

The diagram shows a block of wood in the shape of a prism with triangular cross-section. The end faces are right-angled triangles with sides of lengths $3x$ cm, $4x$ cm and $5x$ cm, and the length of the prism is y cm, as shown in the diagram.



The total surface area of the five faces is 144 cm^2 .

- (a) (i) Show that $xy + x^2 = 12$.

(3)

- (ii) Hence show that the volume of the block, $V \text{ cm}^3$, is given by

$$V = 72x - 6x^3$$

(2)

- (b) (i) Find $\frac{dV}{dx}$.

(2)

- (ii) Show that V has a stationary value when $x = 2$.

(2)

- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 2$.

(2)

(Total 11 marks)

Q26.

A curve C has the equation

$$y = \frac{x^3 + \sqrt{x}}{x}, x > 0$$

- (a) Express $\frac{x^3 + \sqrt{x}}{x}$ in the form $x^p + x^q$.

(3)

- (b) (i) Hence find $\frac{dy}{dx}$.

(2)

- (ii) Find an equation of the normal to the curve C at the point on the curve where

$$x = 1.$$

(4)

(c) (i) Find $\frac{d^2y}{dx^2}$.

(2)

(ii) Hence deduce that the curve C has no maximum points.

(2)

(Total 13 marks)

Q27.

A curve has equation $y = x^3 - 12x$.

The point A on the curve has coordinates $(2, -16)$.

The point B on the curve has x -coordinate $2 + h$.

(a) Show that the gradient of the line AB is $6h + h^2$.

(4)

(b) Explain how the result of part (a) can be used to show that A is a stationary point on the curve.

(2)

(Total 6 marks)

Q28.

A model car moves so that its distance, x centimetres, from a fixed point O after time t seconds is given by

$$x = \frac{1}{2}t^4 - 20t^2 + 66t, \quad 0 \leq t \leq 4$$

(a) Find:

(i) $\frac{dx}{dt}$;

(3)

(ii) $\frac{d^2x}{dt^2}$.

(2)

(b) Verify that x has a stationary value when $t = 3$, and determine whether this stationary value is a maximum value or a minimum value.

(4)

- (c) Find the rate of change of x with respect to t when $t = 1$.

(2)

- (d) Determine whether the distance of the car from O is increasing or decreasing at the instant when $t = 2$.

(2)

(Total 13 marks)

Q29.

A curve has equation $y = e^{2x}(x^2 - 4x - 2)$.

- (a) Find the value of the x -coordinate of each of the stationary points of the curve.

(6)

- (b) (i) Find $\frac{d^2y}{dx^2}$.

(2)

- (ii) Determine the nature of each of the stationary points of the curve.

(2)

(Total 10 marks)

Q30.

The curve with equation $y = x^5 + 20x^2 - 8$ passes through the point P , where $x = -2$.

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Verify that the point P is a stationary point of the curve.

(2)

- (c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P .

(3)

- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)

- (d) Find an equation of the tangent to the curve at the point where $x = 1$.

(4)

(Total 13 marks)

Q31.

A curve has equation

$$y = x^2 - 6x + 5$$

The points A and B on the curve have x -coordinates 2 and $2 + h$ respectively.

- (a) Find, in terms of h , the gradient of the line AB , giving your answer in its simplest form.

(5)

- (b) Explain how the result of part (a) can be used to find the gradient of the curve at A . State the value of this gradient.

(3)

(Total 8 marks)

Q32.

The curve with equation $y = x^4 - 32x + 5$ has a single stationary point, M .

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Hence find the x -coordinate of M .

(3)

- (c) (i) Find $\frac{d^2y}{dx^2}$.

(1)

- (ii) Hence, or otherwise, determine whether M is a maximum or a minimum point.

(2)

- (d) Determine whether the curve is increasing or decreasing at the point on the curve where $x = 0$.

(2)

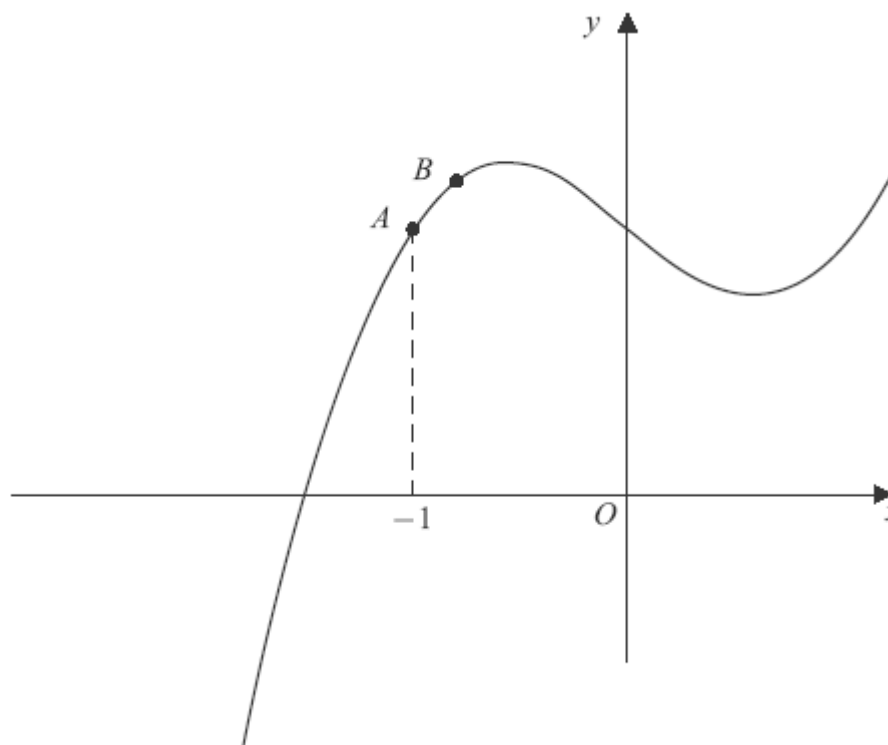
(Total 11 marks)

Q33.

The diagram shows the curve

$$y = x^3 - x + 1$$

The points A and B on the curve have x -coordinates -1 and $-1 + h$ respectively.



- (a) (i) Show that the y -coordinate of the point B is

$$1 + 2h - 3h^2 + h^3$$

(3)

- (ii) Find the gradient of the chord AB in the form

$$p + qh + rh^2$$

where p , q and r are integers.

(3)

- (iii) Explain how your answer to part (a)(ii) can be used to find the gradient of the tangent to the curve at A . State the value of this gradient.

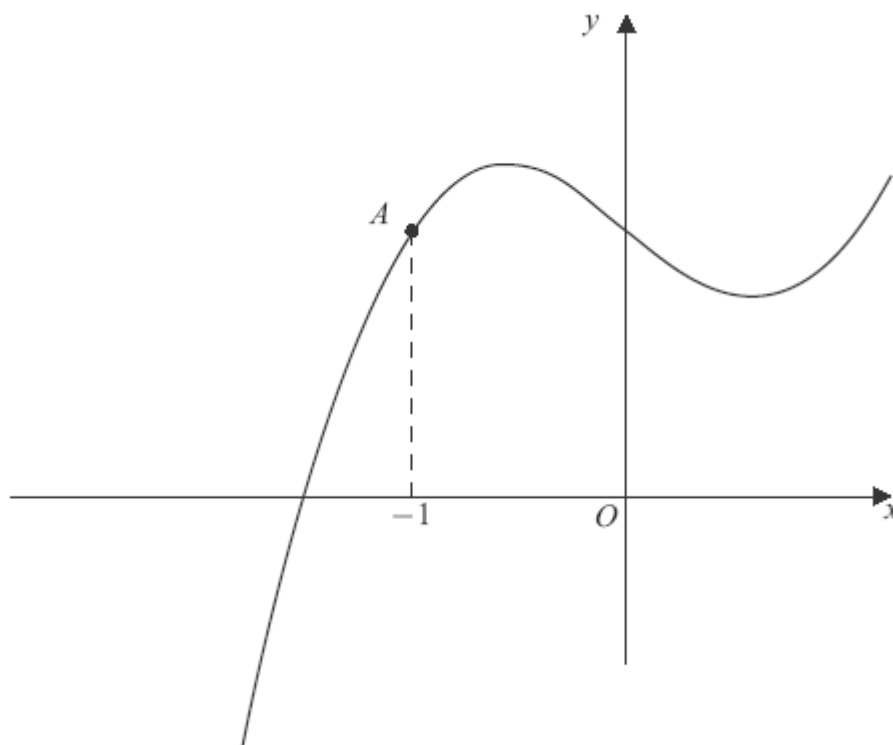
(2)

- (b) The equation $x^3 - x + 1 = 0$ has one real root, α .

- (i) Taking $x_1 = -1$ as a first approximation to α , use the Newton-Raphson method to find a second approximation, x_2 , to α .

(2)

- (ii) On the figure below, draw a straight line to illustrate the Newton-Raphson method as used in part (b)(i). Show the points $(x_2, 0)$ and $(\alpha, 0)$ on your diagram.



(2)

(Total 12 marks)

Q34.

Two numbers, x and y , are such that $3x + y = 9$, where $x \geq 0$ and $y \geq 0$.

It is given that $V = xy^2$.

(a) Show that $V = 81x - 54x^2 + 9x^3$.

(2)

(b) (i) Show that $\frac{dV}{dx} = k(x^2 - 4x + 3)$, and state the value of the integer k .

(4)

(ii) Hence find the two values of x for which $\frac{dV}{dx} = 0$.

(2)

(c) Find $\frac{d^2V}{dx^2}$.

(2)

(d) (i) Find the value of $\frac{d^2V}{dx^2}$ for each of the two values of x found in part (b)(ii).

(1)

(ii) Hence determine the value of x for which V has a maximum value.

(1)

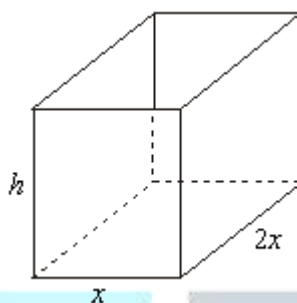
(iii) Find the maximum value of V .

(1)

(Total 13 marks)

Q35.

The diagram shows an **open-topped** water tank with a horizontal rectangular base and four vertical faces. The base has width x metres and length $2x$ metres, and the height of the tank is h metres.



The combined internal surface area of the base and four vertical faces is 54 m^2 .

(a) (i) Show that $x^2 + 3xh = 27$.

(2)

(ii) Hence express h in terms of x .

(1)

(iii) Hence show that the volume of water, $V \text{ m}^3$, that the tank can hold when full is given by

$$V = 18x - \frac{2x^3}{3}$$

(1)

(b) (i) Find $\frac{dV}{dx}$.

(2)

(ii) Verify that V has a stationary value when $x = 3$.

(2)

(c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 3$.

(2)

(Total 10 marks)



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